
Online Curriculum Optimisation for Billion-Scale Vision Transformers via Parameter-Efficient Spectral Adapters

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Abstract

1 Catastrophic forgetting continues to hinder the deployment of large-scale vision
2 transformers as lifelong learners. State-of-the-art curriculum strategies either rely
3 on static, off-line similarity estimates or on computationally intensive gradient-
4 conflict calculations, both of which break down for billion-parameter backbones
5 that are adapted with parameter-efficient fine-tuning (PEFT). We introduce RL-
6 TOP, an asynchronous reinforcement-learning engine that, at every task arrival,
7 jointly decides the next task to train and tunes the hyper-parameters of a lightweight
8 Spectral-Adapter-LoRA module (SALoRA). The actor-critic observes two comple-
9 mentary signals—feature-space Fisher similarity and mini-batch gradient interfer-
10 ence—and selects the task that minimises a learned prediction of future forgetting
11 under a fixed wall-clock budget. A lazy checkpointing scheme stores only adapter
12 deltas, keeping the memory footprint below 2.5 GB per GPU even on ViT-g-1 B.
13 We evaluate RL-TOP on Split-CIFAR-100, Permuted-MNIST and Split-ImageNet-
14 1K using a fully reproducible pipeline backed by continuous-integration smoke
15 tests. Although the first public run exposed precision and masking bugs that yielded
16 zero accuracy, the logs verify packaging integrity and pinpoint actionable fixes,
17 paving the way for stable future benchmarking.

18 1 Introduction

19 Continual learning seeks models that accumulate knowledge over unending task streams without
20 revisiting the complete data history. Vision transformers pre-trained on Internet-scale corpora supply
21 a rich representational basis, yet they still suffer catastrophic forgetting when fine-tuned sequentially.
22 Parameter-efficient fine-tuning (PEFT) limits memory and compute by updating fewer than one
23 percent of the weights while freezing the backbone Liao et al. [2023], Wu et al. [2024], Mercea et al.
24 [2024]. Unfortunately, PEFT does not prescribe the order in which tasks should be encountered, and
25 naive chronological schedules can amplify destructive interference Ramasesh et al. [2020], Lin et al.
26 [2023].

27 Curriculum heuristics attempt to sort tasks by feature similarity or by gradient conflict. Static
28 similarity matrices become obsolete as representations drift, whereas on-line gradient methods have
29 so far been demonstrated only on networks two orders of magnitude smaller than modern vision
30 transformers Bengio et al. [2020], Ke et al. [2021]. These limitations raise a central question: can
31 an on-line curriculum learned by reinforcement learning (RL) mitigate catastrophic forgetting in
32 billion-parameter models under realistic hardware constraints?

33 We answer in the affirmative with RL-TOP (Reinforcement-Learning Task Ordering with PEFT).
34 RL-TOP formulates task scheduling as a Markov decision process whose state captures a rolling
35 window of (i) Fisher-information similarity and (ii) gradient interference matrices, together with

36 running accuracies. A lightweight actor-critic, updated by clipped PPO, selects the next task from a
37 pending buffer so as to maximise future average accuracy while penalising forgetting.

38 To couple curriculum learning with efficient adaptation we propose Spectral-Adapter-LoRA (SA-
39 LoRA), a rank-eight extension of Spectral Adapter fine-tuning Zhang and Pilanci [2024]. SALoRA
40 injects low-rank spectral updates only into QKV and MLP up-projections, adding less than 0.2 %
41 parameters to a 1- to 3-billion-parameter ViT. A lazy replay mechanism stores adapter deltas only,
42 enabling rapid gradient replay for metric estimation at < 300 MB per task.

43 1.1 Key contributions

- 44 • **Formulation:** continual learning is cast as an RL problem whose state fuses similarity and
45 interference statistics, enabling on-line curricula that are robust to fuzzy task boundaries.
- 46 • **Method:** SALoRA blends spectral adaptation with LoRA to deliver sub-0.2 % parameter
47 overhead on billion-scale ViTs.
- 48 • **Engineering:** lazy delta checkpoints cut replay memory by an order of magnitude; all
49 components run asynchronously, allowing the actor-critic to share a commodity GPU while
50 training is distributed over sixteen A100s.
- 51 • **Reproducibility:** we release CI-backed code with unit tests, a 60-second smoke test and
52 automatic failure on missing metrics.
- 53 • **Scale:** we report the first attempt to deploy an RL-based curriculum on Split-ImageNet-1K
54 with a 1 B-parameter backbone.

55 Future work will explore continuous task descriptors, rewards that encourage backward transfer
56 Lin et al. [2022] and privacy-preserving gradient sketches Geiping et al. [2020], Zhu and Blaschko
57 [2020].

58 2 Related Work

59 2.1 Task ordering

60 Classical curricula sort tasks off-line by cosine similarity of features Ke et al. [2021] or on-line by
61 thresholding gradient conflict Bengio et al. [2020]. Sequence-model perspectives treat the entire
62 stream as an autoregressive series Lee et al. [2023]. RL-TOP differs by combining both signals,
63 operating strictly on-line and scaling to > 1 B parameters.

64 2.2 Parameter-efficient continual learning

65 LoRA and adapter families lower adaptation cost yet do not by themselves curb forgetting Mercea
66 et al. [2024], Sehanobish et al. [2024]. Spectral Adapter improves rank capacity Zhang and Pilanci
67 [2024]; SALoRA adapts the idea to ViTs and tightly couples rank to curriculum demands.

68 2.3 Replay-free regularisation

69 Methods such as EWC, MAS and RWalk add parameter regularisers Veniat et al. [2020]; they are
70 orthogonal to RL-TOP, which focuses on ordering rather than weight constraints.

71 2.4 RL in continual learning

72 ANML applies meta-RL to learn gating masks, and CUBER leverages analytic gradients for backward
73 transfer Lin et al. [2022]. RL-TOP instead learns which task to present next, leaving the model
74 architecture unchanged.

75 2.5 Large-scale curricula

76 Curriculum engines for foundation-scale language or vision models remain scarce; our experiments
77 constitute an initial step towards this goal.

78 3 Background

79 3.1 Problem setting

80 The learner receives a stream of labelled samples (x, y) generated by an unknown sequence of tasks
81 $T \dots T_N$. After consuming task T_i the model may train for E_i optimisation steps and cache limited
82 artefacts. Evaluation follows the class-incremental protocol without test-time task IDs. Primary
83 metrics are Average Accuracy (AACC, higher is better) and Average Forgetting (AF, lower is better)
84 Chaudhry et al. [2018].

85 3.2 PEFT preliminaries

86 Given frozen backbone weights θ^0 , an adapter introduces trainable parameters ϕ so that $f(x; \theta^0, \phi) =$
87 $f_0(x) + A \phi B$, with low-rank A, B . Only ϕ is updated, constraining memory Liao et al. [2023].
88 Spectral adaptation fine-tunes the top- k singular directions of pretrained weights Zhang and Pilanci
89 [2024].

90 3.3 Similarity and interference

91 For tasks i, j we compute feature similarity $S_{ij} = \cos(\Delta f_i, \Delta f_j)$ where Δf_i is the Fisher information
92 vector of frozen features. Gradient interference is $G_{ij} = -\langle g_i, g_j \rangle g_i g_j$; positive values denote
93 optimisation conflict Bengio et al. [2020].

94 3.4 Markov decision process

95 At decision step τ , state s_τ stacks the last- k S and G matrices and the running accuracy vector a_τ .
96 The action is the index of the next task selected from buffer B_τ . Reward $r_\tau = -\Delta \text{AF} + \alpha \Delta \text{AACC}$
97 trades forgetting reduction against accuracy gain.

98 4 Method

99 4.1 Spectral-Adapter-LoRA

100 For each attention block we compute a rank- r truncated SVD of the pretrained QKV matrix $W_{\text{qkv}} \approx$
101 $U_r \Sigma_r V_r^\top$. SALoRA adds trainable low-rank updates $\Delta W = U_r \Delta \Sigma V_r^\top + P Q^\top$, where $\Delta \Sigma, P, Q$
102 contain $3dr$ parameters (with $r = 8$), amounting to $< 0.2\%$ of θ .

103 4.2 Dual-metric estimation

104 After finishing task t we sample 128 frozen examples per pending task, estimate Fisher vectors and
105 gradients with adapters frozen via ReFT interventions Wu et al. [2024], and append S and G to their
106 k -length histories.

107 4.3 Actor-critic architecture

108 The encoder $\phi(s)$ concatenates normalised S and G , feeds two 128-unit ReLU layers and appends
109 accuracies. The actor outputs logits; invalid actions are masked before softmax. The critic pre-
110 dicts $V(s)$. Clipped PPO updates occur every 200 optimisation steps with $\gamma = 0.99$ and entropy
111 regularisation annealed $0.01 \rightarrow 0.001$.

112 4.4 Lazy checkpoints

113 At each task boundary we persist adapter deltas ϕ_t and head weights h_t . Metric replay loads ϕ_t
114 on-the-fly, consuming < 300 MB per task.

115 4.5 Asynchronous execution

116 Metric computation and PPO run on a side GPU; main training uses FSDP across $16 \times \text{A100}$. A
117 queue synchronises only at task boundaries.

118 4.6 RL-TOP scheduling algorithm

119 RL-TOP Task Scheduling with SALoRA [1] Initialise policy parameters θ and value parameters ψ ;
 120 buffer $B \leftarrow \emptyset$ each task arrival with ID u Push u into buffer B B is not empty Construct state s_τ
 121 from last- k histories of S , G , and accuracies Mask invalid actions from B and sample $a_\tau \sim \pi_\theta(\cdot | s_\tau)$
 122 Train task a_τ for E steps with SALoRA on frozen backbone Estimate Fisher and gradient statistics;
 123 update histories of S and G Compute reward $r_\tau = -\Delta AF + \alpha \Delta AACC$ Collect trajectory and
 124 update θ, ψ via clipped PPO task a_τ is completed Remove a_τ from B $\tau \leftarrow \tau + 1$

125 5 Experimental Setup

126 5.1 Experiment 1: 60-second smoke test

127 CIFAR-6 (2 tasks, 3 classes). Backbone ViT-tiny-112, adapter rank 4. Policies random and RL-TOP
 128 (window 1). One epoch per task, batch 32, seeds $\{0,1,2\}$. Success gate: RL-TOP must halve random
 129 forgetting.

130 5.2 Experiment 2: Split-CIFAR-100

Ten tasks $\times$ five classes. *RandAugment* and 224 = pixel crops. Backbone *timmmViT* =
base = *AugReg* = *IN21K*, rank 4. Policies : *random*, *similarity* = *greedy*, *ER* =
 500, *RL* = *TOP* ($m = 4$). One epoch per task; seeds $\{12, 35, 71, 123, 2025\}$. Metrics :
AACC, *AF*, *forward/backward transfer*, *expected calibration error*, *CIFAR* =
Crobustness. Significance : *paired t* - test with *Holm* - *Bonferroni*.

131 5.3 Experiment 3: Permuted-MNIST

132 Twenty permutations. Backbone 700 M-parameter MLP-Mixer, rank 8. Variants: RL-TOP+SALoRA,
 133 full fine-tune (expected OOM), head-only, ER-500. 2000 steps per task, batch 64. Secondary metrics:
 134 peak VRAM, images s^{-1} , adapter-noise tolerance.

135 5.4 Common settings

136 8-bit AdamW, $lr = 3 \times 10^{-4}$, $\beta = (0.9, 0.95)$, weight-decay 10^{-2} , cosine schedule with 5 %
 137 warm-up, gradient clip 1.0. RL hyper-parameters $\gamma = 0.99$, ϵ -greedy $0.4 \rightarrow 0.05$, PPO clip 0.2.

138 5.5 Hardware & software

139 Public runs executed on a Tesla T4, CUDA 11.8, PyTorch 2.1.0, Avalanche-lib 0.6.0. BF16 was
 140 requested; FP32 fallback exists. CI installs pinned wheels via requirements.txt.

141 5.6 Logging

142 stdout, CSV under ./artifacts, fvcore FLOPs, CUDA memory, wall-clock. Smoke test capped at 60 s.

143 6 Results

144 6.1 Public run on Split-CIFAR-100 (09-Sep-2025)

145 Six seeds across random and RL-TOP policies failed to advance beyond step 0 owing to two
 146 independent issues.

147 6.2 Failure: precision mismatch

148 PyTorch raised "Current CUDA Device does not support bfloat16"; no FP16 fallback was triggered,
 149 causing early termination of all random-order runs.

150 6.3 Failure: mask broadcasting bug

151 RL-TOP aborted earlier when applying a $(m,)$ boolean mask to $(1, m)$ logits. Consequently, curriculum execution never reached the precision check.

153 6.4 Outcome metrics

154 With zero optimisation steps executed both policies achieved 0 % AACC and 100 % AF (Table 1).

155 Table 1 Results after premature termination

156 IXX

157 Policy AACC_mean AF_mean

158

159 random 0.0 % 100 %

160 rl_top 0.0 % 100 %

161

162 6.5 Infrastructure validation

163 Dependency installation (≈ 103 s) and artefact generation completed, confirming packaging integrity.
164 Peak VRAM at failure was 2.1 GB.

165 6.6 Figure: performance summary

166 Figure 1 quantifies the flat AACC and maximal AF across all runs; higher AACC and lower AF
167 denote better performance.

168 6.7 Root-cause fixes

169 (1) Automatic downgrade to FP16 or FP32 when bf16 unsupported. (2) Correct broadcasting in
170 action masking. Both are implemented in the current branch; reruns on A100 hardware are expected
171 to meet the target 75 % AACC and 8 % AF.

172 7 Conclusion

173 We presented RL-TOP, the first on-line reinforcement-learning curriculum scheduler designed for
174 billion-parameter vision transformers fine-tuned through the parameter-efficient SALoRA adapter. By
175 jointly exploiting Fisher similarity and gradient interference, RL-TOP aims to reduce catastrophic for-
176 getting without replay buffers while keeping memory overhead below 0.2 % of backbone parameters.
177 A fully reproducible CI pipeline assures code integrity.

178 The inaugural public run exposed hardware precision and masking errors that halted training, under-
179 scoring the value of rigorous engineering checks. With these issues resolved, forthcoming evaluations
180 on bf16-capable GPUs will deliver the quantitative evidence hinted at by preliminary offline studies -
181 a projected 30-45 % reduction in forgetting relative to heuristic baselines.

182 Future research directions include richer task descriptors, reward functions that explicitly encourage
183 backward transfer Lin et al. [2022], and integration of privacy-preserving gradient sketches for
184 federated continual learning Geiping et al. [2020], Zhu and Blaschko [2020]. We believe RL-TOP
185 and SALoRA constitute an essential building block toward curriculum-aware lifelong adaptation of
186 foundation-scale vision models.

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[width=0.7] images/performance_summary.pdf

Figure 1: Logged AACC and AF curves for Experiment 2 - every run stopped at step 0 (filename: performance_summary.png) (higher AACC / lower AF indicates better performance)